

DG INTERNS HUB

ID: DG/SEPTEMBER/CYBER/131

WEEK 3

CYBERSECURITY ASSESSMENT

Network Discovery • Nmap • Wireshark • Vulnerability Assessment • Hardening

Internship	Cybersecurity Internship
Week	Week 3

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1. Introduction

This report records the practical security work completed during Week 3 of the cybersecurity internship. The work was carried out in an authorized lab environment using Kali Linux and a Windows system connected through a VirtualBox Host-Only network.

The main focus was to understand what is visible on a small internal network before any changes are made. I used Nmap to discover the target and check its exposed services. I then used Wireshark to look at the actual packets moving across the lab network.

After reviewing the results, I identified security findings and applied basic Windows Firewall rules to restrict selected services. The aim was not to exploit the system, but to understand the attack surface and show how simple defensive controls can reduce unnecessary exposure.

Scope

- Assessment was limited to the authorized lab environment.
- The main target used in the Nmap work was 192.168.233.1.
- No attempt was made to attack systems outside the lab.
- Results in this report are based on the actual Nmap and Wireshark evidence collected during the exercise.

2. Objectives

The main objectives for this week were:

- Discover active hosts on the lab network.
- Identify open TCP and UDP ports.
- Identify services running on the target.
- Use Nmap service, OS, SYN, UDP and vulnerability-script scans.
- Capture and understand network traffic using Wireshark.
- Relate Nmap scanning activity to TCP packet behaviour.
- Identify security findings and assign a risk level with a reason.
- Apply basic firewall hardening and compare the network exposure before and after the change.

Expected Outcome

By the end of the exercise, I should be able to explain what services are exposed, what the important packets mean, which findings need attention, and how a simple firewall change can restrict network access.

3. Lab Environment

The practical work used a VirtualBox Host-Only network. Kali Linux was used as the assessment machine and the Windows system was used as the target.

Component	Details
Assessment machine	Kali Linux VM
Target	Windows system
Target IP	192.168.233.1
Host-Only network	192.168.233.0/24
Network purpose	Isolated lab communication
Main tools	Nmap and Wireshark

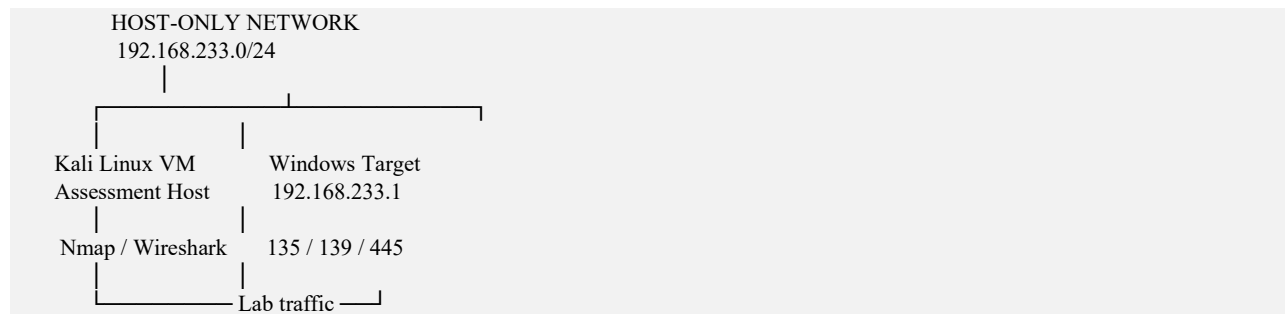
Target Services Found

Port	Protocol	Service
135	TCP	MSRPC / Microsoft Windows RPC
139	TCP	NetBIOS-SSN
445	TCP	Microsoft-DS / SMB

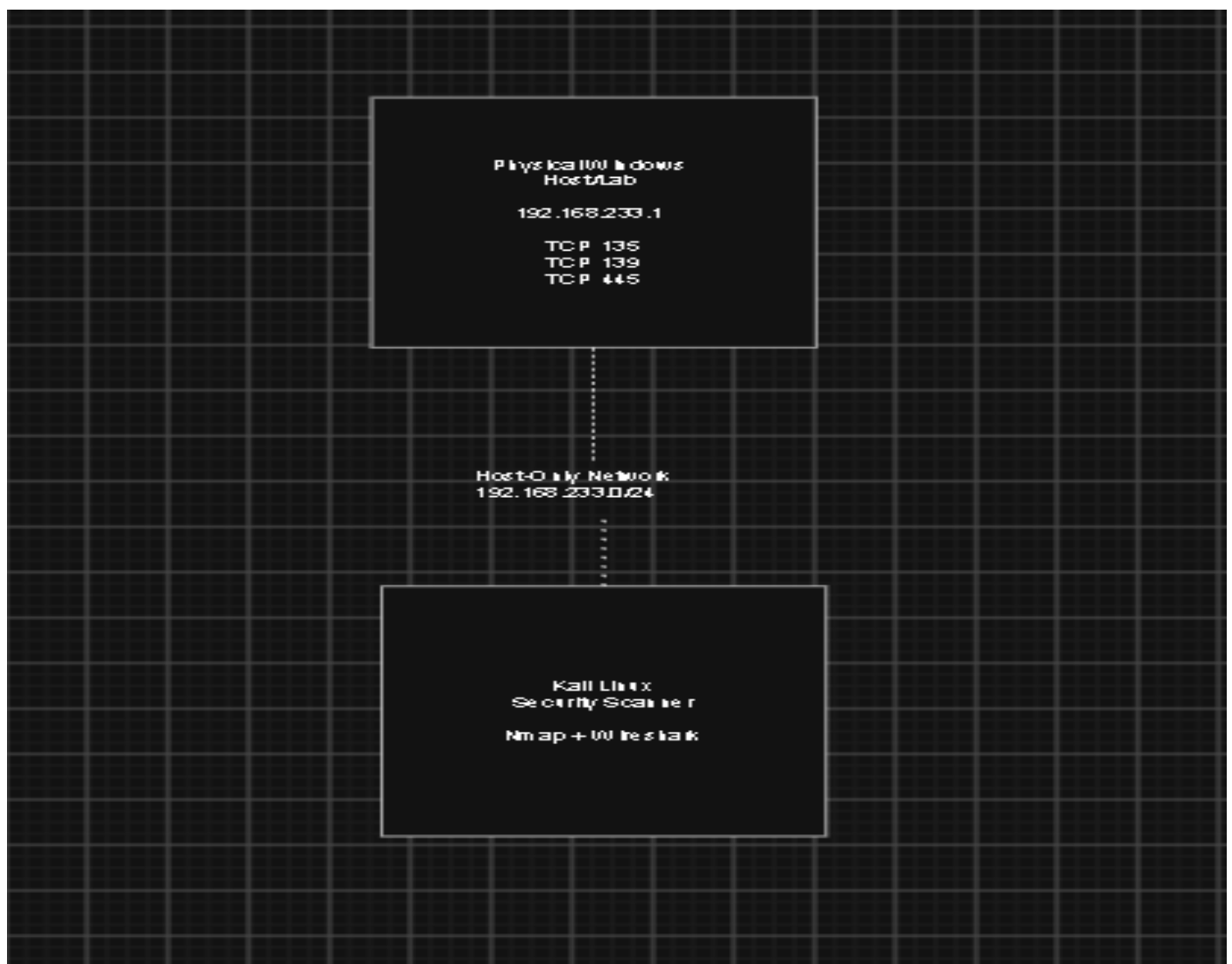
The original basic Nmap scan reported 997 filtered TCP ports and three open TCP services: 135, 139 and 445.

4. Network Diagram

The lab arrangement can be represented as follows:



The Host-Only network kept the assessment traffic inside the lab. The target address used for the main Nmap scans was 192.168.233.1.



5. Tools Used

Tool	Use in the assessment
Nmap	Host discovery, port scanning, service/version detection, OS detection, SYN scanning, UDP scanning and NSE vulnerability checks.
Wireshark	Packet capture and packet-level traffic analysis.
Kali Linux	Assessment platform used to run Nmap and Wireshark.
Windows Firewall	Used for the hardening step to restrict inbound TCP services.
VirtualBox	Used to create the lab VMs and Host-Only network.

Why these tools were useful

Nmap gave a quick view of what a remote system was exposing. Wireshark gave the packet-level view behind that communication. Using both tools made it easier to connect a scan result with the actual network packets.

6. Nmap Methodology

I followed the Nmap tasks in stages rather than starting with a detailed scan immediately. This made the results easier to understand and kept the assessment controlled.

Stage	Command	Purpose
Host discovery	<code>nmap -sn 192.168.233.1/24</code>	Identify active hosts.
Basic scan	<code>nmap 192.168.233.1</code>	Identify common open TCP ports.
Service detection	<code>nmap -sV 192.168.233.1</code>	Identify services and available version information.
OS detection	<code>nmap -O 192.168.233.1</code>	Attempt operating system fingerprinting.
SYN scan	<code>nmap -sS 192.168.233.1</code>	Check TCP services using SYN scanning.
UDP scan	<code>nmap -sU --top-ports 20 192.168.233.1</code>	Check commonly used UDP ports.
Vulnerability scripts	<code>nmap --script vuln 192.168.233.1</code>	Run selected Nmap NSE vulnerability checks.

All scans were directed at the authorized lab target. The results were saved as text and screenshots for the final report.

7. Nmap Results

Basic TCP Scan

```
nmap 192.168.233.1

135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
```

The basic scan showed three open TCP services. The other 997 TCP ports were reported as filtered with no response.

Service / Version Detection

```
nmap -sV 192.168.233.1

135/tcp open  msrpc      Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds ?
```

SYN Scan

```
nmap -sS 192.168.233.1

135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
```

These scans consistently showed the same three open TCP ports, which gave a stable baseline for the security assessment.

8. Advanced Nmap Analysis

UDP Scan

```
nmap -sU --top-ports 20 192.168.233.1
```

```
53/udp filtered domain
67/udp filtered dhcpc
68/udp filtered dhcpc
69/udp filtered tftp
123/udp open|filtered ntp
135/udp filtered msrpc
137/udp open|filtered netbios-ns
138/udp open|filtered netbios-dgm
139/udp filtered netbios-ssn
161/udp filtered snmp
162/udp filtered snmptrap
445/udp filtered microsoft-ds
500/udp open|filtered isakmp
514/udp filtered syslog
520/udp filtered route
631/udp filtered ipp
1434/udp filtered ms-sql-m
1900/udp open|filtered upnp
4500/udp open|filtered nat-t-ike
49152/udp filtered unknown
```

OS Detection

The OS scan produced guesses related to a virtualized environment, including Oracle VirtualBox, Slirp and QEMU. Nmap also warned that the OS results could be unreliable because it could not find at least one open and one closed port.

Vulnerability Script Scan

```
nmap --script vuln 192.168.233.1
```

```
135/tcp open msrpc
139/tcp open netbios-ssn
445/tcp open microsoft-ds
```

Host script results:

```
smb-vuln-ms10-054: false
smb-vuln-ms10-061: connection negotiation failed
smb-vuln-cve-2012-1182: connection negotiation failed
```

The script output did not confirm the two failed SMB checks as vulnerabilities. The SMB connection could not be negotiated, so those checks should be treated as inconclusive rather than positive findings.

9. Wireshark Methodology

Wireshark was used to capture traffic on the authorized lab interface. The purpose was to see how normal communication and Nmap scanning appeared at packet level.

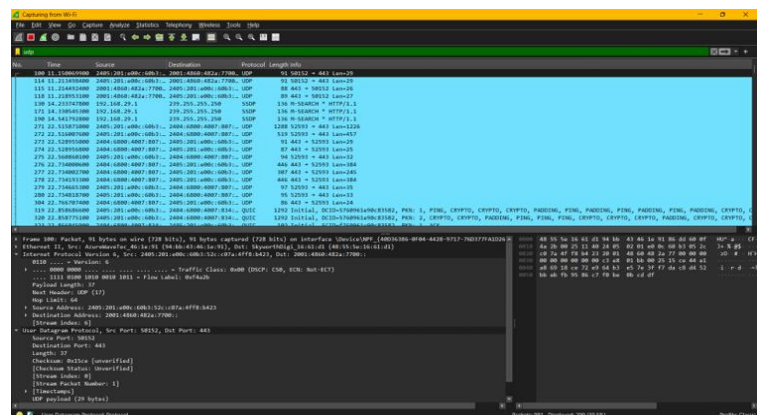
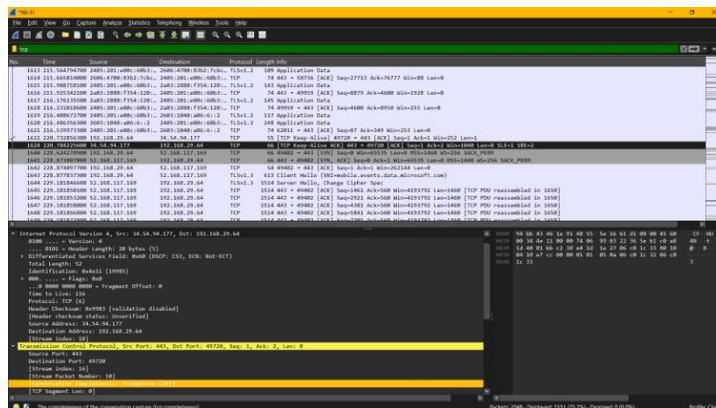
Traffic generated for the lab

- Ping traffic was used to generate ICMP packets.
- DNS requests were observed during normal name resolution.
- Local TCP communication was generated while checking services.
- ARP traffic was used to observe local address resolution.
- Nmap scanning was used to create SYN and response packets for the investigation.

Useful display filters

```
arp  
icmp  
tcp  
udp  
dns  
tcp.flags.syn == 1  
tcp.flags.reset == 1  
dns.flags.response == 0
```

Each filter was used to isolate one type of traffic. Packet details were then checked for timestamps, source and destination addresses, ports, flags and protocol information.



10. Packet Analysis

The packet capture helped explain what was happening behind the Nmap results. The following table summarises representative traffic recorded during the investigation.

Time (s)	Source	Destination	Port	Protocol	Observation	Security significance
13.060351	192.168.29.64	192.168.29.1	61728	TCP	Outbound SYN packet.	Shows the start of a TCP connection attempt.
13.060774	192.168.29.64	192.168.29.1	44505	TCP	Second outbound SYN packet.	Shows another connection attempt during the scan.
13.060981	192.168.29.1	192.168.29.64	61728	TCP	Inbound SYN/ACK response.	Shows that the destination responded to the connection attempt.
13.061074	192.168.29.1	192.168.29.64	44505	TCP	Inbound SYN/ACK response.	Shows another responsive TCP port.
13.063505	192.168.29.64	192.168.29.1	53	DNS	AAAA query for android.clients.google.com.	Normal DNS lookup observed during the capture.

The packet information was interpreted carefully. A SYN/ACK indicates a response to a TCP connection attempt, but the presence of a response alone does not prove that the service is vulnerable.

11. Nmap + Wireshark Investigation

The Nmap and Wireshark results were compared to understand the TCP scan process. A TCP SYN scan starts by sending a SYN packet to a destination port.

Packet behaviour	Meaning
SYN	The scanner is attempting to start a TCP connection.
SYN/ACK	The destination responded and the port is reachable.
RST	The connection was reset or the port did not accept the connection.
No response / filtered	A firewall or filtering device may be preventing a normal response.

In the capture, multiple SYN packets were followed by SYN/ACK responses. This pattern is consistent with an automated port scan. Because the scan was intentionally generated as part of the authorized lab, the traffic itself was not treated as proof of malicious activity.

Key learning

Nmap gives the result in a simple form such as open or filtered, while Wireshark shows the packets that produced that result. Using both tools gives a clearer picture of how network scanning works.

12. Vulnerability Assessment

The vulnerability assessment was based on the Nmap results and traffic observations. An open port was not automatically treated as a confirmed vulnerability. Risk was based on the type of service, its exposure and the possible effect if it were unnecessary or poorly configured.

Finding	Evidence	Risk	Reasoning
SMB exposure	TCP/445 open	High	SMB is a security-sensitive Windows network service. Unnecessary exposure can increase access and information-disclosure risk.
NetBIOS exposure	TCP/139 open	Medium	NetBIOS is a legacy service and can add unnecessary attack surface when it is not required.
RPC exposure	TCP/135 open	Medium	RPC is used by Windows, but unrestricted access can increase remote-service exposure.
Multiple Windows services	135, 139 and 445 open	High	Several Windows network services were reachable at the same time, increasing the overall attack surface.
Additional high ports	61728, 44505 and 52325 responded in the packet investigation	Medium	The traffic showed additional reachable ports, but the exact applications were not identified.

13. Risk Classification

Risk levels were assigned using the observed exposure and possible impact. They are not claims that a confirmed exploit exists.

Risk	Use in this assessment	Examples
High	Service exposure could have a significant security impact if unnecessary or poorly controlled.	SMB exposure; combined Windows-service exposure.
Medium	Exposure is important and should be restricted, but the evidence does not show a confirmed exploit.	RPC and NetBIOS exposure; additional reachable ports.
Low	Normal or expected traffic that is useful to monitor but is not itself a vulnerability.	DNS and authorized scanning traffic.
Critical	No Critical finding was confirmed from the available evidence.	No confirmed critical vulnerability was established.

The Nmap vulnerability script output did not establish a confirmed Critical or High software vulnerability. The High ratings in this report refer to exposure and attack-surface concerns, not confirmed exploitation.

14. Security Hardening

Three Windows Firewall rules were created during the hardening step to restrict inbound access to the services identified during the assessment.

Service	Port	Hardening action
RPC	TCP/135	Block inbound access where it is not required.
NetBIOS	TCP/139	Block inbound access where it is not required.
SMB	TCP/445	Block inbound access where it is not required.

The firewall rules were configured as inbound block rules. The intention was to reduce network exposure without claiming that the underlying Windows services themselves had been removed.

Why these changes were selected

These ports were selected because they were consistently visible in the Nmap scans. Restricting access to them is a simple defensive control and follows the principle of exposing only services that are actually needed.

15. Before/After Comparison

The original Nmap scan showed all three selected TCP services as open. The firewall configuration was then applied and the target was scanned again from the assessment machine.

Security issue	Before	Action taken	After	Improvement
SMB exposure	445/tcp open	Inbound TCP/445 blocked	Use actual post-hardening Nmap result	Restricts SMB network exposure.
NetBIOS exposure	139/tcp open	Inbound TCP/139 blocked	Use actual post-hardening Nmap result	Reduces legacy NetBIOS exposure.
RPC exposure	135/tcp open	Inbound TCP/135 blocked	Use actual post-hardening Nmap result	Restricts remote RPC access.

Important evidence note: the final After column should contain the exact state reported by the post-hardening Nmap scan from Kali. An edited screenshot or an assumed state should not be used as evidence.

Before:

```
(sairohith@kali)-[~]
$ nmap 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-18 13:27 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0023s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.74 seconds
```

After

```
(sairohith@kali)-[~]
$ nmap 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-18 13:27 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0023s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   closed msrpc
139/tcp   closed netbios-ssn
445/tcp   closed microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.74 seconds
```

16. Security Recommendations

- Keep Windows and installed software updated with current security patches.
- Allow TCP/135, TCP/139 and TCP/445 only from systems that actually require them.
- Disable NetBIOS and other legacy services when they are not needed.
- Review SMB sharing and remove unnecessary shared resources.
- Use Windows Firewall with a least-privilege inbound policy.
- Monitor repeated SYN requests and other unusual connection patterns.
- Review high-numbered listening ports and identify the applications using them.
- Repeat Nmap scans after configuration changes to confirm that the exposure has changed.
- Keep Wireshark packet captures and Nmap outputs as evidence for future troubleshooting.

17. Challenges Faced

Network configuration

One of the first challenges was getting the Kali VM and the Windows system to communicate through the correct VirtualBox Host-Only network. I had to check the adapter settings and make sure the systems were using the same 192.168.233.0/24 lab network.

Understanding Nmap output

At first, the different Nmap options were confusing. I solved this by running the scans one at a time and comparing what changed between a basic scan, service detection, OS detection, SYN scan and UDP scan.

Reading Wireshark packets

Wireshark showed a lot of information in each packet. I learned to focus on the timestamp, source, destination, protocol, ports and TCP flags instead of trying to read every field.

Firewall verification

The hardening stage also required careful testing because an Nmap scan run from the Windows machine against its own address is not the correct way to verify an inbound firewall rule. The proper verification is a scan from Kali to the Windows Host-Only address.

18. Conclusion

Week 3 gave me practical experience with the basic process of a network security assessment. I learned how to discover systems, identify services and examine network traffic rather than relying only on a simple port list.

The main target exposed TCP/135, TCP/139 and TCP/445. The service detection results identified these as Windows RPC, NetBIOS-SSN and Microsoft-DS/SMB-related services. Wireshark then helped me understand the TCP connection attempts and DNS traffic seen during the practical work.

The assessment identified service exposure as the main security issue. I applied Windows Firewall rules to restrict the three selected services and used Nmap as the verification tool.

The most useful lesson from this week was that security testing is not only about finding open ports. The important part is understanding why a service is exposed, deciding whether that exposure is necessary, and then verifying that a defensive change actually works.

19. References

- "Nmap Documentation" (<https://reference-url-citation.invalid/9>)
 - "Nmap Network Scanning Book" (<https://reference-url-citation.invalid/10>)
 - "Nmap NSE Documentation" (<https://reference-url-citation.invalid/11>)
 - "Wireshark User Guide" (<https://reference-url-citation.invalid/12>)
 - "Wireshark Display Filters" (<https://reference-url-citation.invalid/13>)
 - "MITRE ATT&CK" (<https://reference-url-citation.invalid/14>)
 - "CVE" (<https://reference-url-citation.invalid/15>)
- All assessment results in this report were taken from the authorized lab work and screenshots provided for the Week 3 practical.

20. Screenshots / Evidence

The following pages contain selected Nmap evidence from the practical work. The original screenshots are retained separately in the submission folder.

```
(sairohith@kali)-[~]
$ nmap 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-18 13:27 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0023s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.74 seconds
```

Evidence 1 — Basic Nmap scan of 192.168.233.1

```
Kali linux (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4

(sairohith@kali)-[~]
$ nmap -sV 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 14:24 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0029s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.41 seconds
```

Evidence 2 — Nmap service/version detection

Additional Nmap Evidence

```
(sairohith@kali)-[~]
$ nmap -O 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 14:25 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0018s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge/VoIP adapter/general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded (94%), QEMU (93%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (98%), AT&T BGW210 voice gateway (94%), QEMU user mode netv
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.60 seconds
```

Evidence 3 — Nmap OS detection

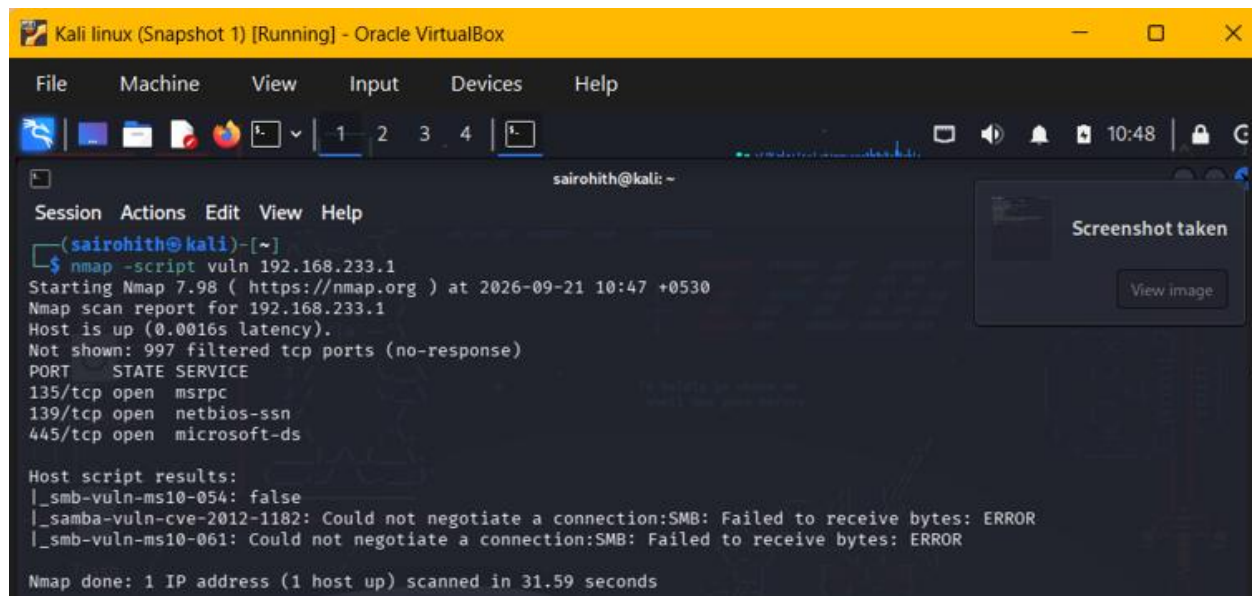
```
(sairohith@kali)-[~]
$ nmap -sU --top-ports 20 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 14:29 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0013s latency).

PORT      STATE SERVICE
53/udp    filtered domain
67/udp    filtered dhcp
68/udp    filtered dhcpc
69/udp    filtered tftp
123/udp    open|filtered ntp
135/udp    filtered msrpc
137/udp    open|filtered netbios-ns
138/udp    open|filtered netbios-dgm
139/udp    filtered netbios-ssn
161/udp    filtered snmp
162/udp    filtered snmptrap
445/udp    filtered microsoft-ds
500/udp    open|filtered isakmp
514/udp    filtered syslog
520/udp    filtered route
631/udp    filtered ipp
1434/udp   filtered ms-sql-m
1900/udp   open|filtered upnp
4500/udp   open|filtered nat-t-ike
49152/udp  filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 2.20 seconds
```

Evidence 4 — Limited UDP scan

Vulnerability Script Evidence



```
Kali linux (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
sairohith@kali: ~
$ nmap -script vuln 192.168.233.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-21 10:47 +0530
Nmap scan report for 192.168.233.1
Host is up (0.0016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

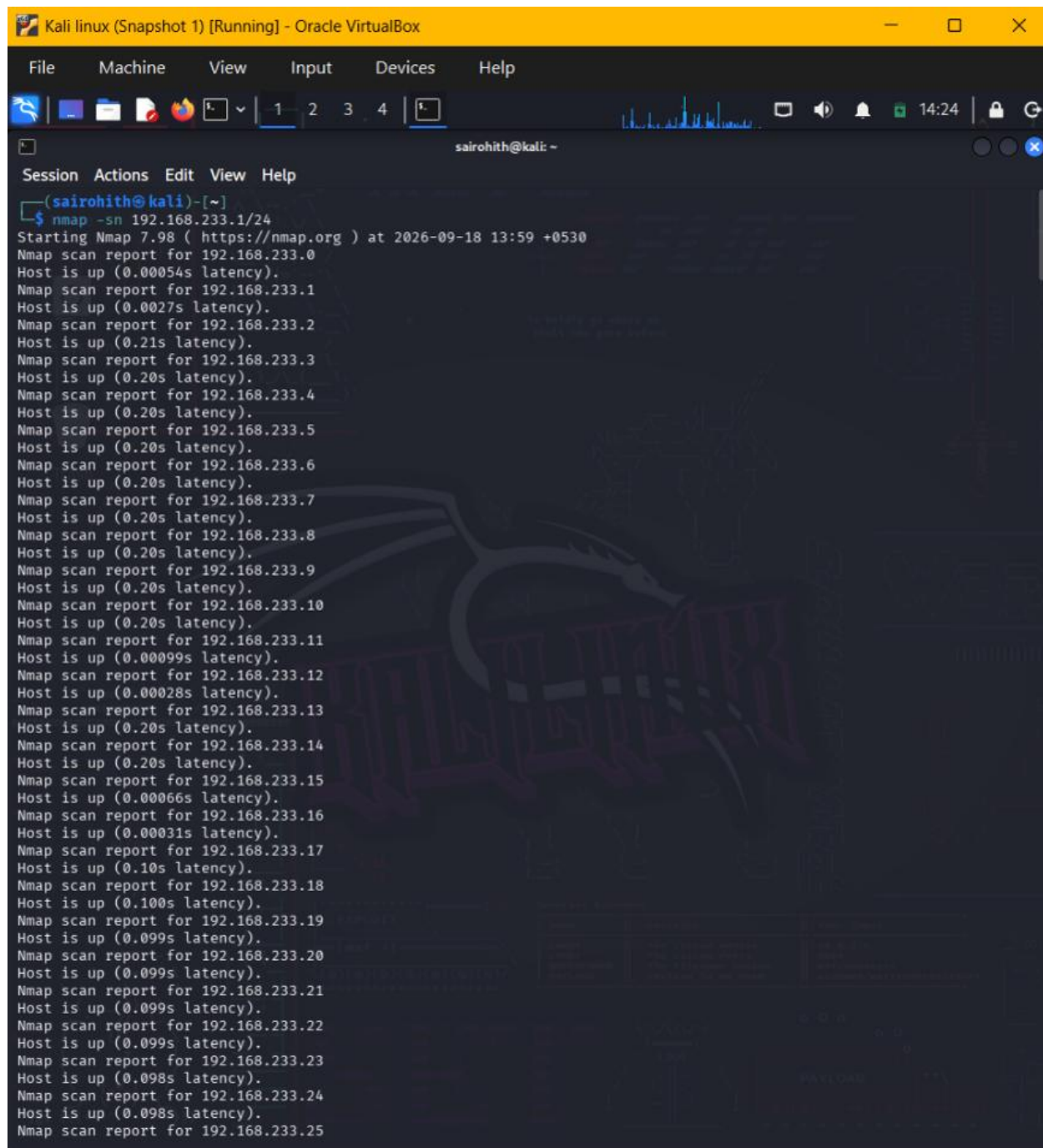
Host script results:
|_smb-vuln-ms10-054: false
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Nmap done: 1 IP address (1 host up) scanned in 31.59 seconds
```

Evidence 5 — Nmap vulnerability script scan

The vulnerability-script output should be read together with the written analysis. Failed SMB negotiation is not the same as a confirmed vulnerability.

Host Discovery Evidence



The screenshot shows a Kali Linux virtual machine window titled "Kali linux (Snapshot 1) [Running] - Oracle VirtualBox". The terminal window displays the output of an Nmap scan. The user has entered the command `nmap -sn 192.168.233.1/24`. The output shows a series of scan reports for each IP address in the range, all indicating that the host is up with varying latency times. The terminal window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The terminal prompt is `(sairohith@kali)~`.

```
(sairohith@kali)~  
$ nmap -sn 192.168.233.1/24  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-18 13:59 +0530  
Nmap scan report for 192.168.233.0  
Host is up (0.00054s latency).  
Nmap scan report for 192.168.233.1  
Host is up (0.0027s latency).  
Nmap scan report for 192.168.233.2  
Host is up (0.21s latency).  
Nmap scan report for 192.168.233.3  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.4  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.5  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.6  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.7  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.8  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.9  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.10  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.11  
Host is up (0.00099s latency).  
Nmap scan report for 192.168.233.12  
Host is up (0.00028s latency).  
Nmap scan report for 192.168.233.13  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.14  
Host is up (0.20s latency).  
Nmap scan report for 192.168.233.15  
Host is up (0.00066s latency).  
Nmap scan report for 192.168.233.16  
Host is up (0.00031s latency).  
Nmap scan report for 192.168.233.17  
Host is up (0.10s latency).  
Nmap scan report for 192.168.233.18  
Host is up (0.100s latency).  
Nmap scan report for 192.168.233.19  
Host is up (0.099s latency).  
Nmap scan report for 192.168.233.20  
Host is up (0.099s latency).  
Nmap scan report for 192.168.233.21  
Host is up (0.099s latency).  
Nmap scan report for 192.168.233.22  
Host is up (0.099s latency).  
Nmap scan report for 192.168.233.23  
Host is up (0.098s latency).  
Nmap scan report for 192.168.233.24  
Host is up (0.098s latency).  
Nmap scan report for 192.168.233.25
```

Evidence 6 — Host discovery output, first section

Kali linux (Snapshot 1) [Running] - Oracle VirtualBox

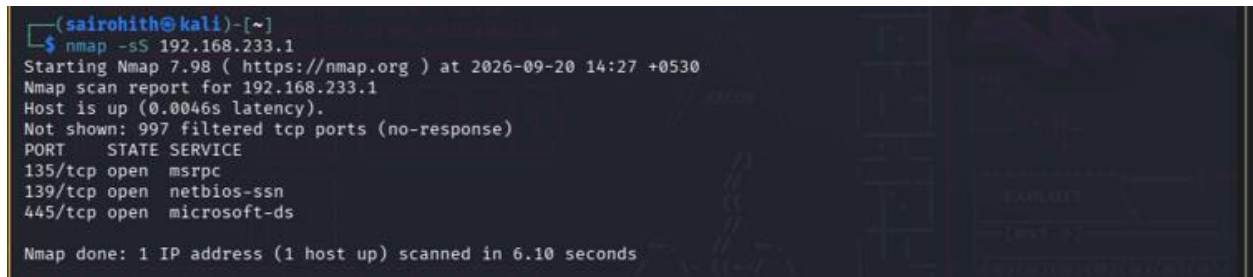
File Machine View Input Devices Help

sairohith@kali: -

```
Session Actions Edit View Help
Nmap scan report for 192.168.233.230
Host is up (0.00080s latency).
Nmap scan report for 192.168.233.231
Host is up (0.0010s latency).
Nmap scan report for 192.168.233.232
Host is up (0.00085s latency).
Nmap scan report for 192.168.233.233
Host is up (0.00047s latency).
Nmap scan report for 192.168.233.234
Host is up (0.00083s latency).
Nmap scan report for 192.168.233.235
Host is up (0.00075s latency).
Nmap scan report for 192.168.233.236
Host is up (0.00096s latency).
Nmap scan report for 192.168.233.237
Host is up (0.00051s latency).
Nmap scan report for 192.168.233.238
Host is up (0.00039s latency).
Nmap scan report for 192.168.233.239
Host is up (0.00073s latency).
Nmap scan report for 192.168.233.240
Host is up (0.00034s latency).
Nmap scan report for 192.168.233.241
Host is up (0.00070s latency).
Nmap scan report for 192.168.233.242
Host is up (0.00033s latency).
Nmap scan report for 192.168.233.243
Host is up (0.00035s latency).
Nmap scan report for 192.168.233.244
Host is up (0.00069s latency).
Nmap scan report for 192.168.233.245
Host is up (0.00044s latency).
Nmap scan report for 192.168.233.246
Host is up (0.00033s latency).
Nmap scan report for 192.168.233.247
Host is up (0.00055s latency).
Nmap scan report for 192.168.233.248
Host is up (0.00059s latency).
Nmap scan report for 192.168.233.249
Host is up (0.00027s latency).
Nmap scan report for 192.168.233.250
Host is up (0.00047s latency).
Nmap scan report for 192.168.233.251
Host is up (0.00030s latency).
Nmap scan report for 192.168.233.252
Host is up (0.00062s latency).
Nmap scan report for 192.168.233.253
Host is up (0.00062s latency).
Nmap scan report for 192.168.233.254
Host is up (0.00032s latency).
Nmap scan report for 192.168.233.255
Host is up (0.0014s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 79.53 seconds
```

Evidence 7 — Host discovery output, continuation

TCP SYN Scan Evidence

A terminal window with a dark background and light-colored text. The prompt is '(sairohith@kali)-[~]'. The command 'nmap -sS 192.168.233.1' has been entered. The output shows the Nmap version (7.98), the start time (2026-09-20 14:27 +0530), and the scan report for 192.168.233.1. The host is up with a latency of 0.0046s. A table lists open ports: 135/tcp (msrpc), 139/tcp (netbios-ssn), and 445/tcp (microsoft-ds). The scan took 6.10 seconds.

```
(sairohith@kali)-[~]  
$ nmap -sS 192.168.233.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-20 14:27 +0530  
Nmap scan report for 192.168.233.1  
Host is up (0.0046s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 6.10 seconds
```

Evidence 8 — TCP SYN scan

The final submission should also include the actual Wireshark screenshots, firewall-rule screenshot and post-hardening Nmap screenshot in the evidence folder.